

HAT-P-37b

R-2452

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-37_181020 · created 2026-08-23T00:08:59+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458411.6196 +/- 0.0046 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.164 +/- 0.049
Transit depth (Rp/Rs)^2	2.68 +/- 1.6 [%]
Orbital Inclination (inc)	84.54 +/- 2.26
Airmass coefficient 1 (a1)	1.052 +/- 0.037
Airmass coefficient 2 (a2)	-0.031 +/- 0.063
Scatter in the residuals of the lightcurve fit is	3.48 %
Best Comparison Star	#3 - [208, 406]
Optimal Aperture	3.26
Optimal Annulus	11.53
Transit Duration (day)	0.059 +/- 0.03

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.164 $\pm$ 0.049 vs 0.1394 (deviation 0.5 σ)
Duration fit vs published	0.059 d vs 0.0983 d (deviation 1.31 σ)
Mid-time O–C	11.9 min
Depth SNR	3.3
Residual scatter	3.48 %

Light curve

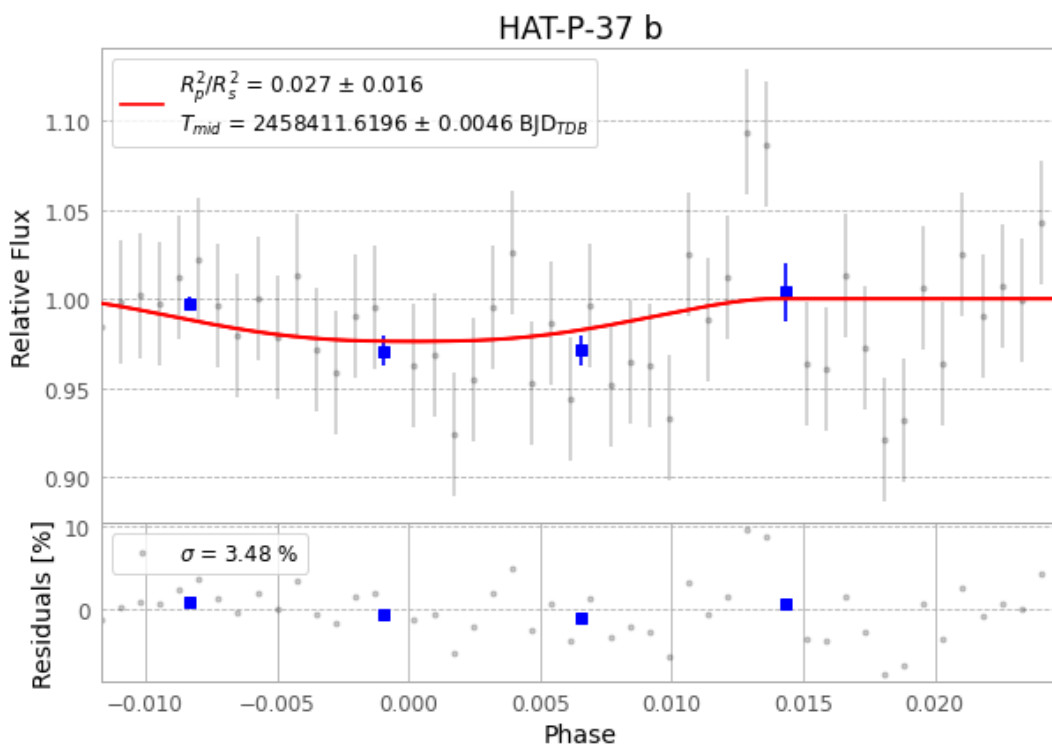


Figure 1 — FinalLightCurve_HAT-P-37 b_2018-10-19.png (SHA-256 3e6401d2b7ec999e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [339, 346], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 4dca3a50d9ee8f12...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

50 FITS frames (33 MB), HATP-37181020020312.FITS → HATP-37181020043012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-37-181020.log (6059fc1085f1...), FinalLightCurve_HAT-P-37 b_2018-10-19.png (3e6401d2b7ec...), FinalLightCurve_HAT-P-37 b_2018-10-19.csv (d55172e05892...)

Compute resources

Wall time 523.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-23 00:09Z.